

Hands-On Labs - AH Coding & MySQL Training

AH200: Advanced CODING & MySQL Integration

OOP, APIs, and MySQL Integration

CODING & MySQL Integration

Objective: Implement CODING scripts to interact with a MySQL database, reinforcing database connectivity, CRUD operations, and data retrieval.

Instructions:

1. Database Creation:

- o Using MySQL, create a database named student_records.

o **Reference:** [Creating MySQL Database and Tables: Step-by-Step Guide](#)

2. Table Creation:

- o Within the student_records database, create a table called students with the following columns:

- id (INT, Primary Key, Auto Increment)
- name (VARCHAR(100))
- age (INT)
- email (VARCHAR(100))

o **Reference:** [MySQL Exercises. Practice. Solution](#)

3. Data Insertion:

- o Insert at least three records into the students table.

o **Reference:** [MySQL Exercises. Practice. Solution](#)

4. Data Retrieval:

- o Write a CODING script to connect to the MySQL database and retrieve all records from the students table. Display the records in an HTML table.

o **Reference:** [CODING Exercises, Practice Questions and Solutions](#)

5. Data Update:

- o Implement a CODING script that updates a student's email based on

their id. o **Reference:** [MySQL Exercises, Practice, Solution](#)

6. Data Deletion:

o Create a CODING script to delete a student record based on their id.

o **Reference:** [MySQL Exercises, Practice, Solution](#)

Expected Outcome:

- Students will understand how to perform CRUD (Create, Read, Update, Delete) operations using CODING and MySQL.

These hands-on labs are designed to reinforce the theoretical knowledge gained during the classes by providing practical experience with CODING and MySQL.

Flask api that allows CRUD operations on a student database. Api end points have been tested with postman

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File Edit Selection View Go Q api

EXPLORER

- API
 - __pycache__
 - instance
 - models
 - __pycache__
 - __init__.cpython-311.pyc
 - student_model.cpython-311.pyc
 - user_model.cpython-311.pyc
 - __init__.py
 - student_model.py
 - routes
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- app.py
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- extensions.py
- students.db

OUTLINE

TIMELINE

users.py config.py U student_model.py U app.py U extensions.py U X students.p

extensions.py > ...

```
1 # extensions.py
2 from flask_sqlalchemy import SQLAlchemy
3 from flask_bcrypt import Bcrypt
4
5 db = SQLAlchemy()
6 bcrypt = Bcrypt()
7
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS python3

(venv)~[elvis@kali:~/Desktop/africa hackon/api]

\$ python3 app.py

IP	Time	Method	Path	Status	Message
127.0.0.1	[14/Dec/2025 14:19:57]	POST	/api/students HTTP/1.1	400	-
127.0.0.1	[14/Dec/2025 14:21:36]	POST	/api/students HTTP/1.1	400	-
127.0.0.1	[14/Dec/2025 14:22:40]	POST	/api/students HTTP/1.1	400	-
127.0.0.1	[14/Dec/2025 14:22:51]	POST	/api/students HTTP/1.1	400	-
127.0.0.1	[14/Dec/2025 14:23:05]	POST	/api/students HTTP/1.1	400	-
127.0.0.1	[14/Dec/2025 14:23:08]	POST	/api/students HTTP/1.1	400	-
127.0.0.1	[14/Dec/2025 14:23:14]	POST	/api/students HTTP/1.1	400	-
127.0.0.1	[14/Dec/2025 14:23:29]	POST	/api/students HTTP/1.1	400	-
127.0.0.1	[14/Dec/2025 14:24:11]	POST	/api/students HTTP/1.1	201	-
127.0.0.1	[14/Dec/2025 14:24:35]	POST	/api/students HTTP/1.1	201	-

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OUTLINE

TIMELINE

users.py config.py U student_model.py U X app.py U students.db U students.py

models > student_model.py > ...

```
1 # Import the database object
2 from extensions import db
3
4 # Define students table
5 class Student(db.Model):
6     # Primary key column
7     id = db.Column(db.Integer, primary_key=True)
8
9     # Student name (max 100 characters)
10    name = db.Column(db.String(100), nullable=False)
11
12    # Student age
13    age = db.Column(db.Integer, nullable=False)
14
15    # Student email (unique)
16    email = db.Column(db.String(100), unique=True, nullable=False)
17
```

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- OUTLINE
- TIMELINE

routes > students.py > ...

```
1 from flask import Blueprint, request, jsonify
2 from models.student_model import Student
3 from extensions import db
4
5 students_bp = Blueprint("students", __name__)
6
7 # .....
8 # CREATE STUDENT
9 # .....
10 @students_bp.route("/students", methods=["POST"])
11 def create_student():
12     data = request.get_json()
13
14     name = data.get("name")
15     age = data.get("age")
16     email = data.get("email")
17
18     if not name or not age or not email:
19         return jsonify({"error": "All fields required"}), 400
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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

python3

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- OUTLINE
- TIMELINE

app.py > create_app

```
1 from flask import Flask
2 from models.student_model import Student
3 from models.user_model import User
4 from extensions import db
5
6 def create_app():
7     app = Flask(__name__)
8     app.config.from_object(Config)
9
10     db.init_app(app)
11
12     # REGISTER STUDENT ROUTES
13     app.register_blueprint(students_bp, url_prefix="/api")
14
15     # CREATE TABLES
16     with app.app_context():
17         db.create_all()
18
19     return app
20
21 if __name__ == "__main__":
22     app = create_app()
23     app.run(debug=True)
```

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